Rubrics Quiz 1 CS310 2025

September 3, 2025

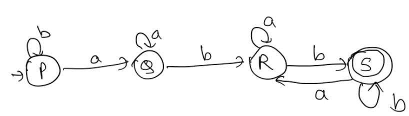
Depending on the question set, one can change Σ accordingly by switching {a,b,c,...} to {0,1,2,...}

Q1

(a) 8 marks
8 - Correct answer & explanation(57 for length 7 and 120 for length 8)
6 - Correct answer, no/wrong explanation
4 - Correct direction of thinking
1 - Reasonable attempt
0 - Otherwise

(b) 7 marks
Answer- $(a + b)^*ab(a + b)^*b$
7- Correct expression and justification
5- Correct expression, no/wrong justification, too long expression
2 to 3 - Reasonable attempt
0 - Otherwise

(c) 10 marks
10- 4 States and Correct with some justification



6- Correct states, 1-2 missing/wrong transitions
4- More than 4 states and almost correct
2- Reasonable attempt
0- Otherwise

Q2

(a) 13 marks
13- Correct CFG with ≤ 8 variables

4 cases

$n<m$ $\quad n>m$ $\quad m>k$ $\quad m<k$

$C^* \rightarrow \varepsilon \mid cC^*$ $\qquad A^* \rightarrow \varepsilon \mid aA^*$

$C^+ \rightarrow cC^*$ $\qquad A^+ \rightarrow aA^*$

$B^+ \rightarrow b \mid bB^+$ $\qquad S_{ab} \rightarrow \varepsilon \mid aS_{ab}b$

$S_{bc} \rightarrow \varepsilon \mid bS_{bc}c$

$S \rightarrow S_{ab}B^+C^*$ $\qquad (n<m$ case$)$

$S \rightarrow A^+S_{ab}C^*$ $\qquad (n>m$ case$)$

$S \rightarrow A^*S_{bc}C^+$ $\qquad (m<k$ case$)$

$S \rightarrow A^*B^+S_{bc}$ $\qquad (m>k$ case$)$

Variables $\rightarrow \{S, S_{ab}, S_{bc}, A^+, A^*, B^+, C^+, C^*\}$

8- Correct CFG with ≤ 12 variables
6- Almost correct CFG with 8 variables
4- Some idea of the 4 cases being reflected in the CFG
2- Reasonable attempt
0- Otherwise

(b) 4 marks
4- Correct justification about the 4 cases to take
2- Some reasonable attempt
0- Otherwise

(c) 8 marks
8- Correct trees
4- Some flaws in the tree
1- Reasonable attempt
0- Otherwise

Q3

(a) 13 marks
13- Fully Correct (Can add sink)
6 to 8- ϵ not accepting rest correct/ minor mistake/ more than 6 states not including sink
2 to 4- Reasonable attempt

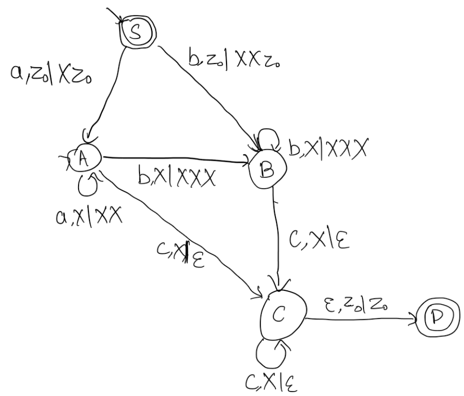


Figure 1: One possible solution to Q3(a)

0- Otherwise

(b) 5 marks
5- Correct run
3 to 4- Minor mistakes
1- Reasonable attempt
0- Otherwise

(c) 7 marks
7- Correct run
5- Minor mistakes
2- Reasonable attempt
0- Otherwise

Q4

(a) 15 marks
15- Correct grammar with ≤ 3 variables and ≤ 4 rules
One such grammar is-

$$\boxed{\begin{array}{r} S- > a|XSY \\ Xa- > aaX \\ XY- > \epsilon \end{array}}$$

10- Correct grammar with ≤ 6 variables and ≤ 10 rules
4 to 6- Almost correct grammar with ≤ 8 variables and ≤ 10 rules
2- Reasonable attempt
0- Otherwise

(b) 5 marks
5- Correct explanation
3 to 4- Some flaws
1- Reasonable attempt
0- Otherwise

(c) 5 marks
5- Correct derivation
3 to 4- Some minor mistakes, skipping intermediate derivation
1 to 2- Reasonable attempt
0- Otherwise